Om Bharadwaj

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PROFESSIONAL SUMMARY

Computer Science and Engineering undergraduate at IIIT Manipur (CGPA 9.05, Top 4%) with production-grade experience in full-stack development, applied machine learning, and cloud-native engineering. Interned at Tata Technologies delivering enterprise CRM infrastructure using React.js, PHP, and MySQL under agile workflows. Independently architected end-to-end AI systems including a medical-imaging CNN ensemble with clinical explainability (Grad-CAM, LIME, SHAP) and a renewable energy forecasting ML pipeline, both containerized and deployed as REST APIs. Proficient in Python, JavaScript, SQL, FastAPI, Node.js, Docker, PyTorch, and scikit-learn, with strong foundations in data structures, system design, and database management. Leads Google Developer Group chapter serving 1200+ students. Seeking software engineering, applied AI, or backend roles to build scalable, high-impact solutions.

EDUCATION

Indian Institute of Information Technology Manipur — Imphal, India
B.Tech in Computer Science & Engineering (AI & Data Science) — 2023 – 2027 (Expected)
CGPA: 9.05 / 10.0 | Top 4% of Batch

EXPERIENCE

Software Engineering Intern — May 2026 – July 2026
Tata Technologies — Remote

- Developed an enterprise CRM Portal in an agile team to streamline customer relationship management workflows for large-scale clients
- Built a responsive React.js frontend with dynamic dashboards, data visualization components, and real-time notification modules
- Engineered secure REST APIs in PHP with role-based access control (RBAC), authentication middleware, and automated report generation
- Designed optimized MySQL schemas using normalized tables, indexed queries, and stored procedures for scalable data operations

TECHNICAL SKILLS

Languages: Python, JavaScript, C++, Java, SQL, PHP, C
Frontend & Backend: React.js, FastAPI, Express.js, Node.js, PHP, REST APIs, Microservices
Databases: MySQL, PostgreSQL, MongoDB, SQLite
AI / ML: PyTorch, scikit-learn, EfficientNet-B4, ResNet-18, GoogLeNet, DenseNet-121, XGBoost, LightGBM, Random Forest, SVM, LIME, SHAP, Grad-CAM, Gemini API, Groq API, Pandas, NumPy, Matplotlib
DevOps & Tools: Docker, Git, GitHub, CI/CD, Render
CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design

PROJECTS

Diabetic Retinopathy Detection with Explainable AI — GitHub
Python · PyTorch · EfficientNet-B4 · ResNet-18 · GoogLeNet · DenseNet-121 · SVM · FastAPI · React.js · Docker

- Architected a hybrid CNN ensemble + SVM pipeline on the APTOS 2019 dataset for binary and five-class retinopathy severity grading; applied five-view TTA and PDR class-weight boosting to address severe class imbalance across severity grades
- Integrated Grad-CAM, LIME, and SHAP for localized saliency maps and clinical interpretability; automated structured patient reports via Gemini and Groq LLM APIs with feature caching to eliminate redundant CNN inference
- Deployed end-to-end as a containerized application (FastAPI + Express.js + React.js + Docker) for reproducible clinical inference

AI-Powered Solar Power Output Prediction — GitHub
Python · XGBoost · LightGBM · Random Forest · FastAPI · Express.js · React.js · Docker · Pandas · NumPy

- Engineered time-aware features (hour-of-day, seasonality, solar angle proxies) from meteorological inputs (irradiance, temperature, humidity, cloud cover, wind speed) to model temporal energy generation patterns
- Trained and benchmarked Random Forest, XGBoost, and LightGBM using cross-validated grid search; selected optimal model by RMSE and R^2 metrics, outperforming all baseline models
- Containerized the full stack with Docker and deployed as a REST API (FastAPI + Express.js backend, React.js frontend) for reproducible production inference

ACHIEVEMENTS & LEADERSHIP

- Achieved CGPA 9.05 at IIIT Manipur across five semesters, placing in the Top 4% of a 228-student cohort
- **GDG On Campus Chapter Lead, IIIT Manipur** — Selected through Google's competitive process evaluating technical depth and community leadership; organizes Study Jams, workshops, and developer events for 1200+ students
- **GeeksforGeeks Campus Ambassador** — Connects students with technical resources, competitive programming initiatives, and industry learning opportunities on campus